

EE6347 - Course Project [Jul-Nov 25]

Team Members:

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*Note: All the required files are included in the folder "Files" in the zip provided.
High Quality images also included in the folder "images"*

Software training & Quantisation: [Aksshith Golla]

Python Notebook for this: *EE6347_TermProject_070_008.ipynb*

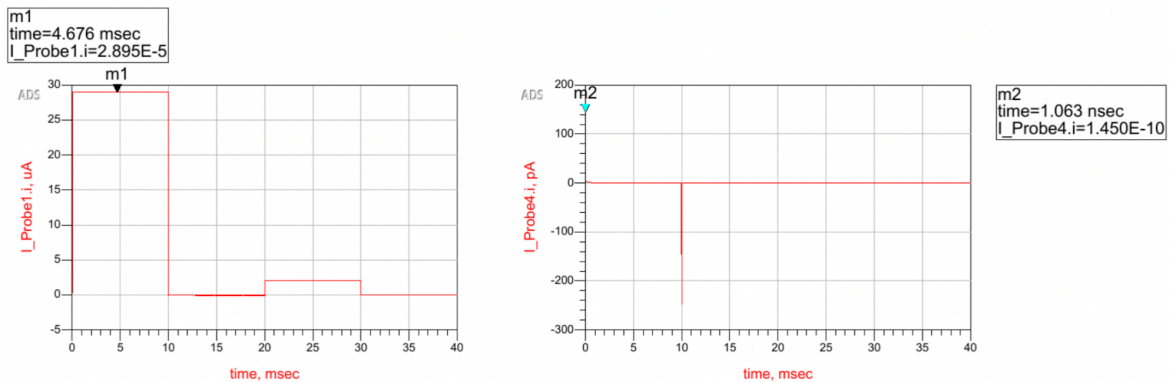
Training accuracy pre-quantisation = **98.44 %**

Training accuracy post-quantisation = **62.81 %**

Synaptic Array: [Himanshu Borkar]

Programming scheme:-

- 1. Program mode:** Weights for a selected row: 0V for 0 bit and 3.0V for 1 bit.
WordLine: 0.9V for selected row/
- 2. Inference mode:** BitLine: Input activations as -0.2V for 0 pixel and 0.2V for 1 pixel.
WordLine: All rows at 0.9V.

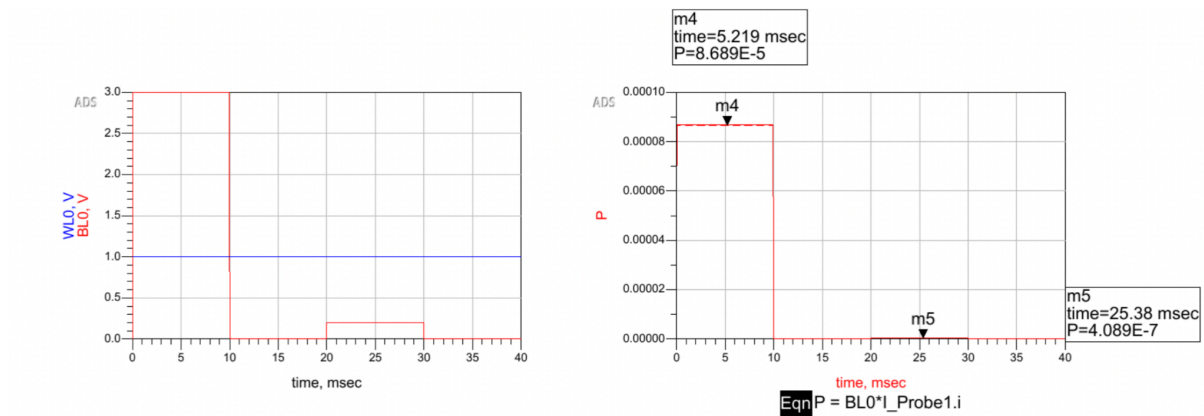


$$I_{ON} = 28.96 \text{ uA}$$

$$I_{OFF} = 14.50 \text{ nA}$$

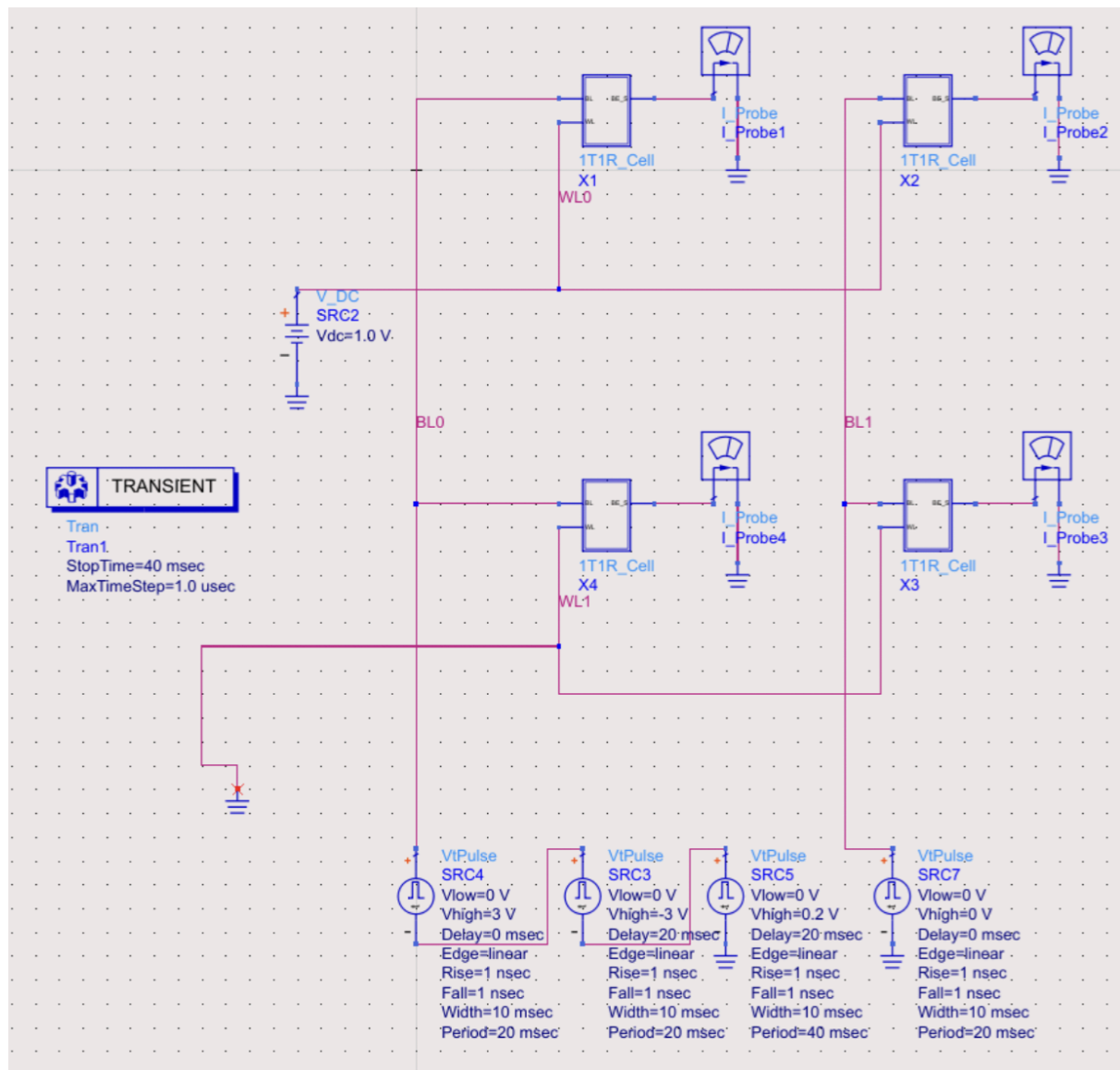
Hence the ON/OFF ratio is:

$$I_{ON}/I_{OFF} = 1997.25$$



Plots for Input (left) and Power consumption (right)

Power consumed in writing a cell is = **86.88 μ W**

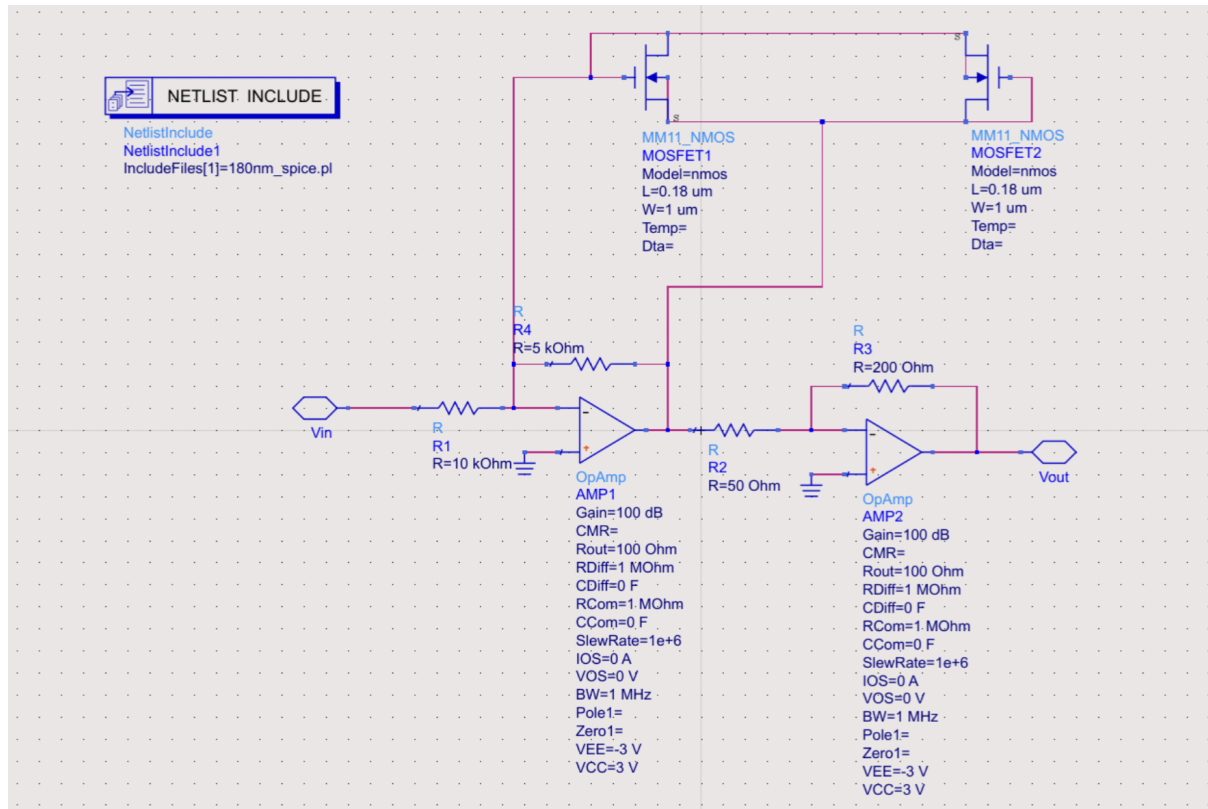


Schematic for ON/OFF ratio and Program power consumption

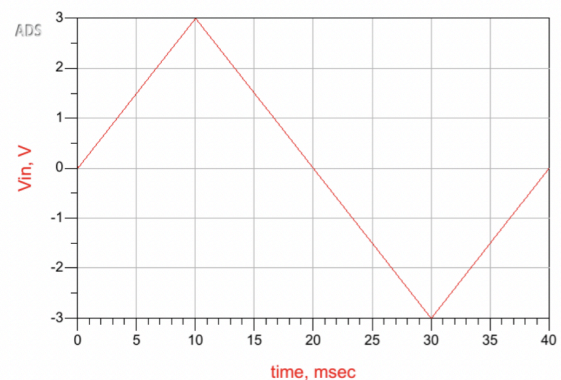
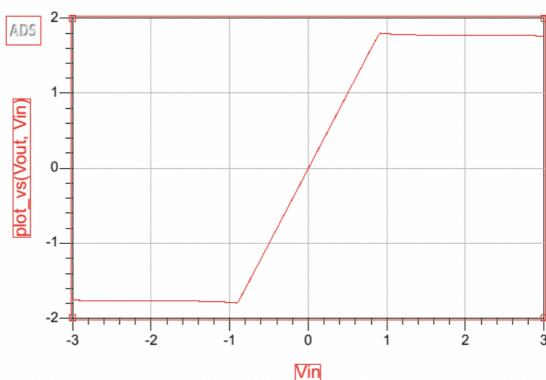
Activation Function & MUXing: [Himanshu & Aksshith]

Activation function: $\tanh$

For implementing the tanh function, we referred to the following source: [1]

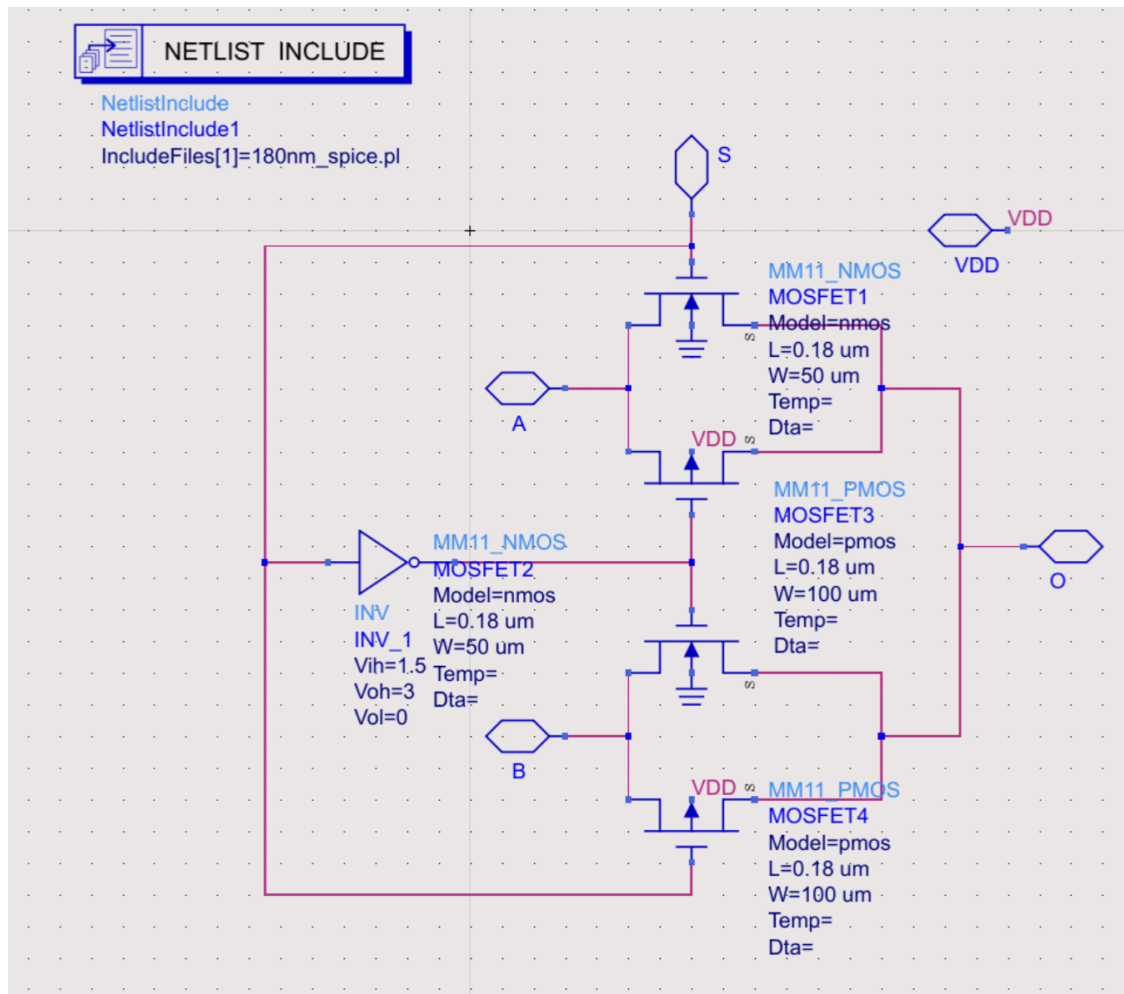


Schematic for CMOS implementation of tanh function



Tanh Function (Approximated) (left) and Input V_{in} (right)

2:1 Analog MUX:

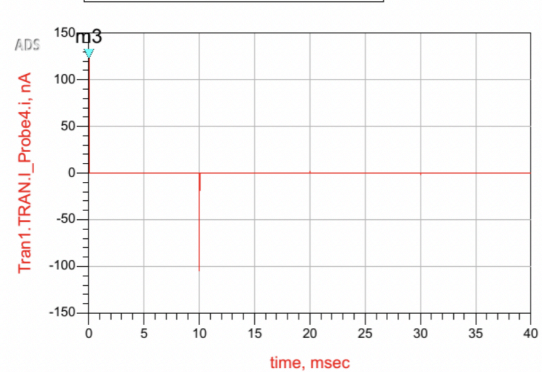
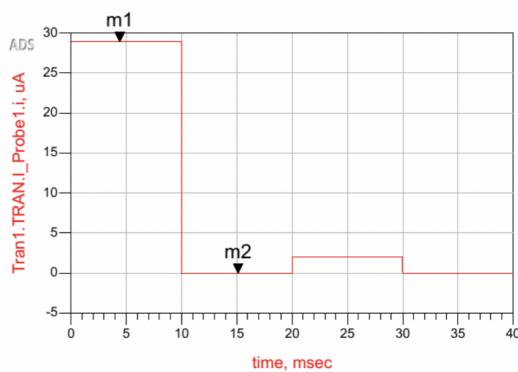


Schematic for the Transmission Gate 2:1 MUX

m2
time=15.10 msec
Tran1.TRAN.I_Probe1.i=6.447E-10

m1
time=4.392 msec
Tran1.TRAN.I_Probe1.i=2.896E-5

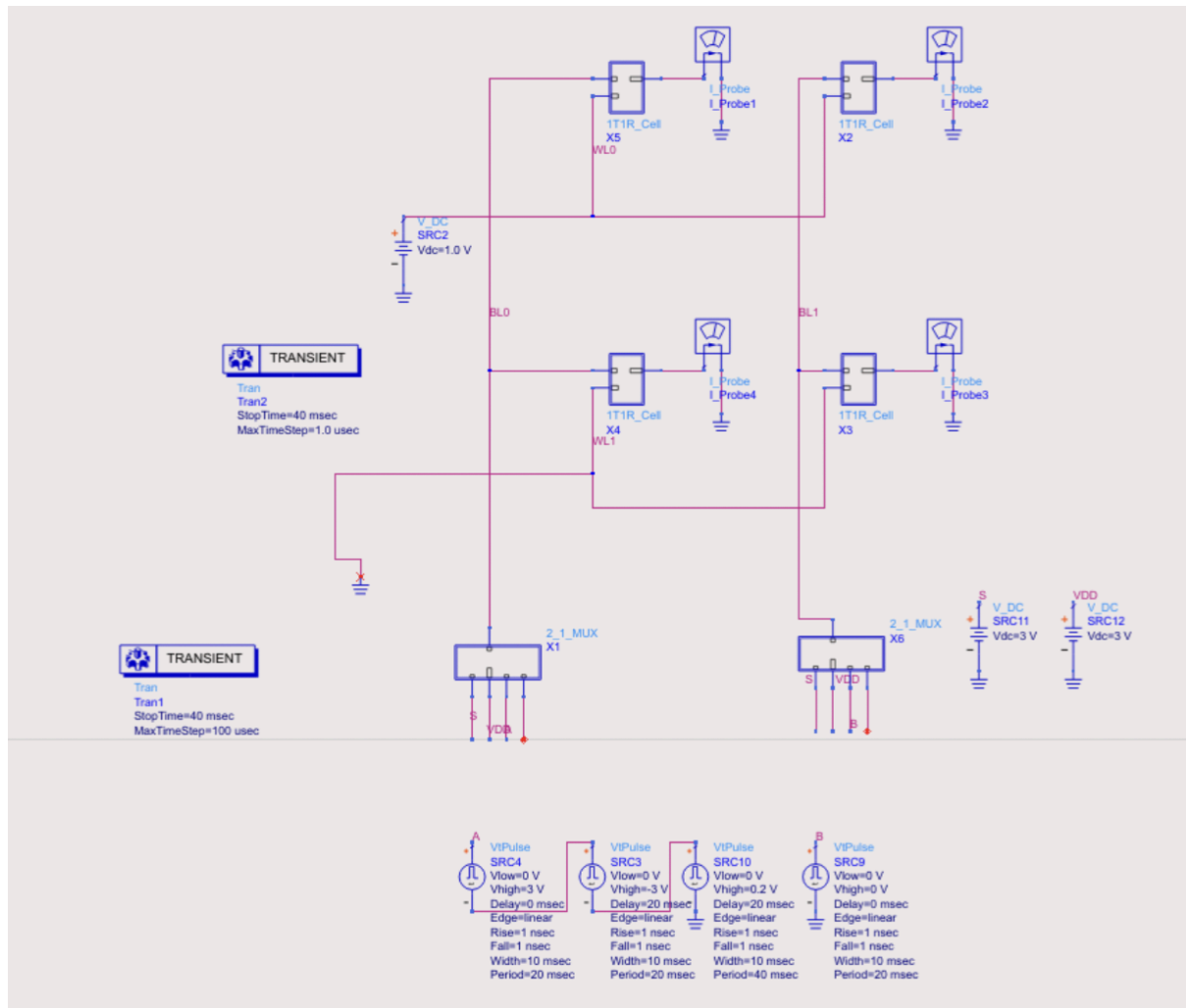
m3
time=1.000 nsec
Tran1.TRAN.I_Probe4.i=1.229E-7



Post 2:1 Mux addition power analysis

I_{ON} after mux = **28.96 uA**

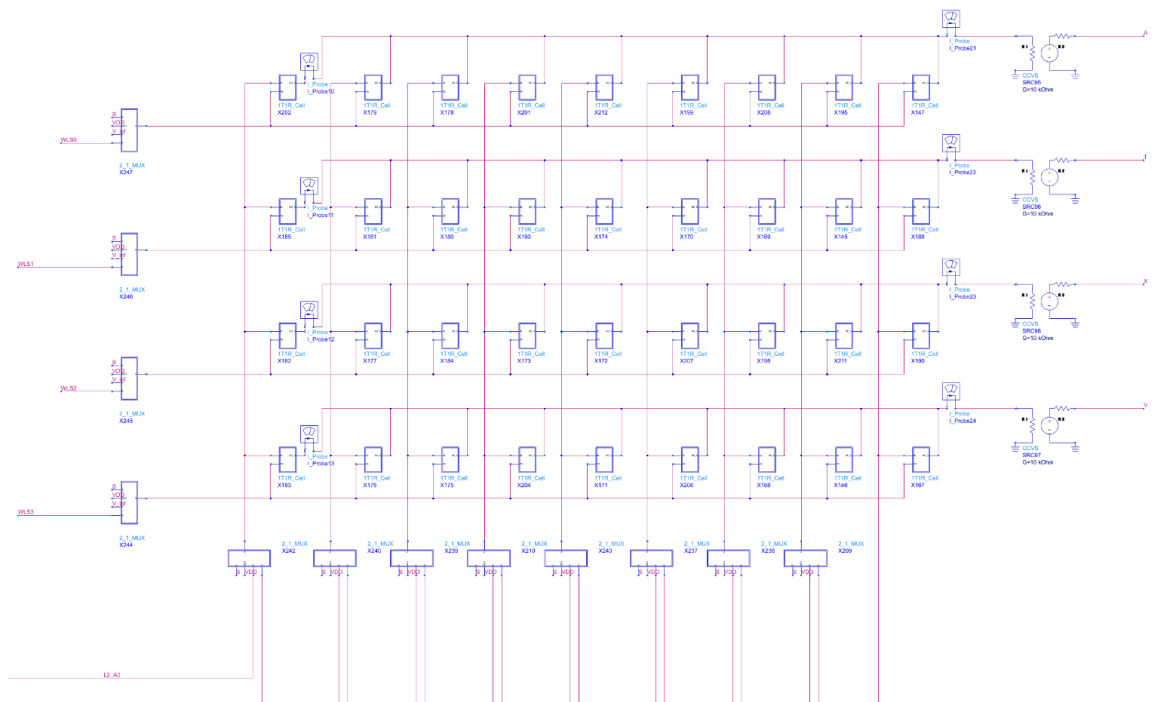
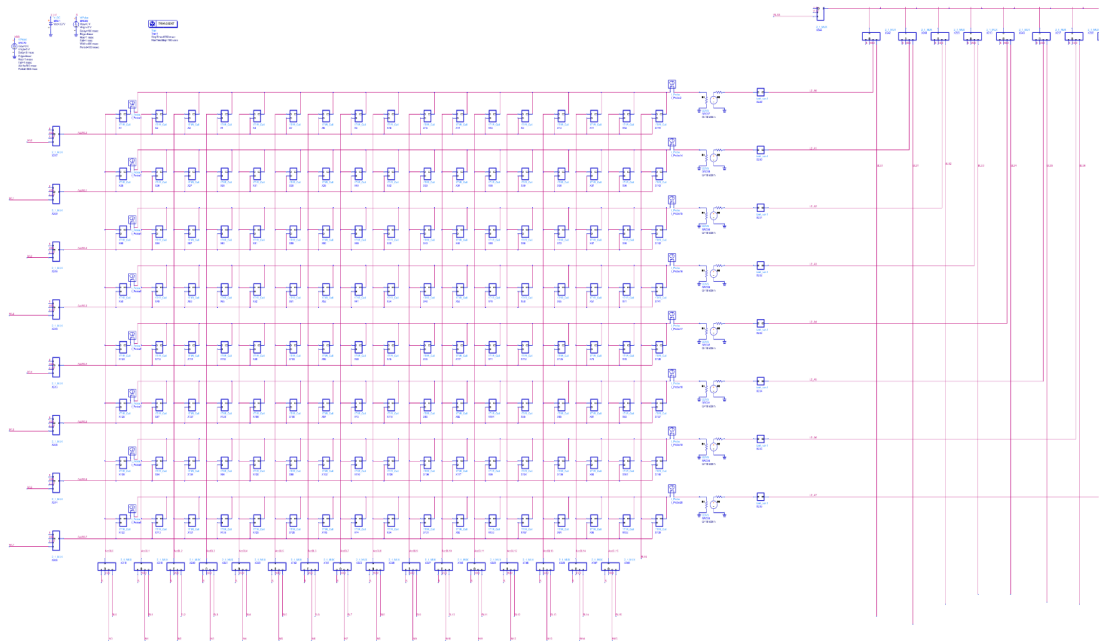
I_{OFF} after mux = **122.9 nA** $\Rightarrow I_{\text{ON}}/I_{\text{OFF}}$ = **235.63** (reduced)



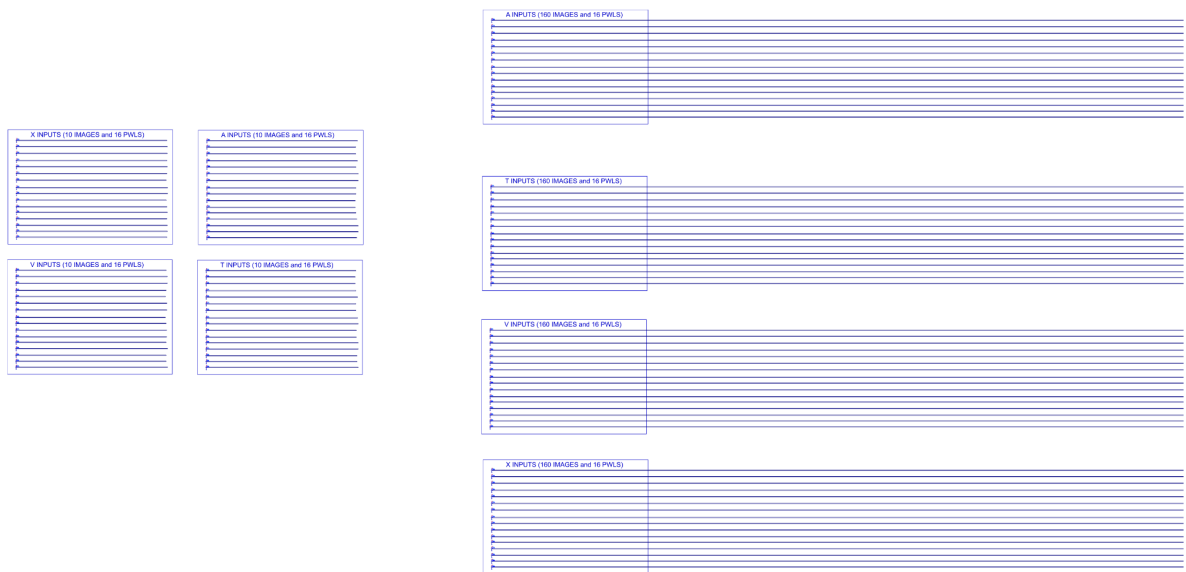
Schematic for ON/OFF Ratio - MUX

Testbench: [Himanshu Borkar]

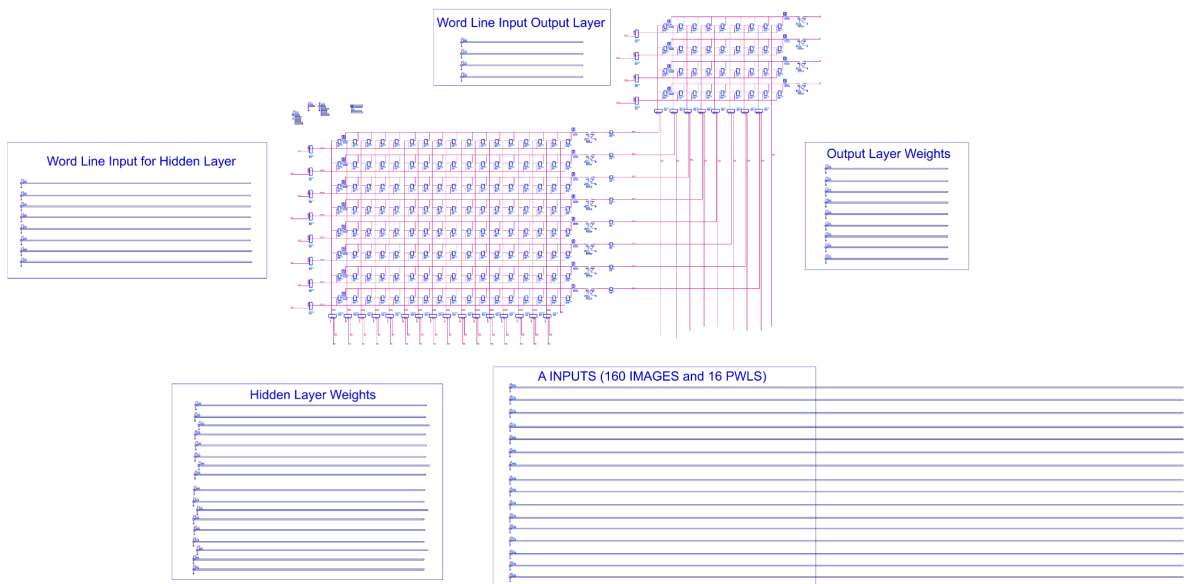
17x8 crossbar for hidden layer with 8 neurons



9x4 crossbar array for output layer



All PWL inputs (zip file has attached image with better quality)



FULL IMPLEMENTATION

Benchmarking: [Himanshu Borkar]

All the images for each batch of inferences are included in the zip file as:

A_infplot.png
T_infplot.png
V_infplot.png
X_infplot.png

Accuracy:

```
Accuracy over 640 images = 24.53125 %
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Scatter Plot:

(code included in zip file in last part of Accuracy_EE6347.py)



ADDITIONALLY: (FFNN - Feedforward Neural Network)

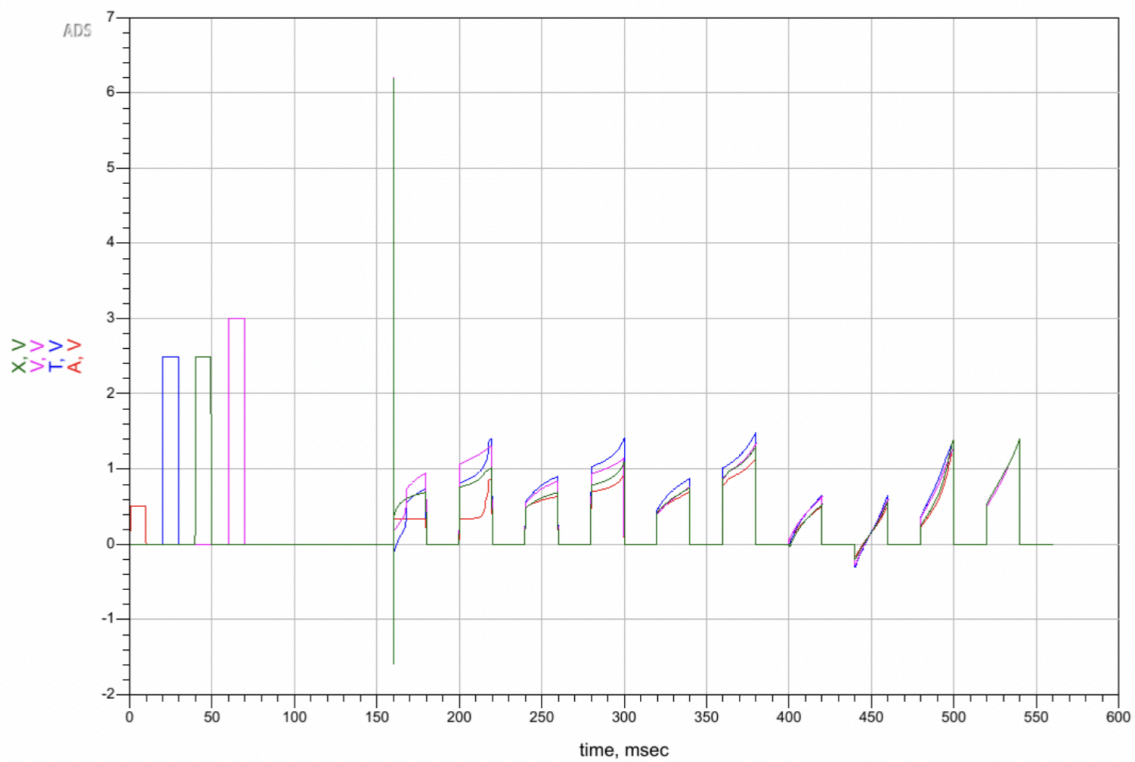


Fig. (TOP) A LETTER Training Data (10 images) as inputs to the 2-layer FFNN

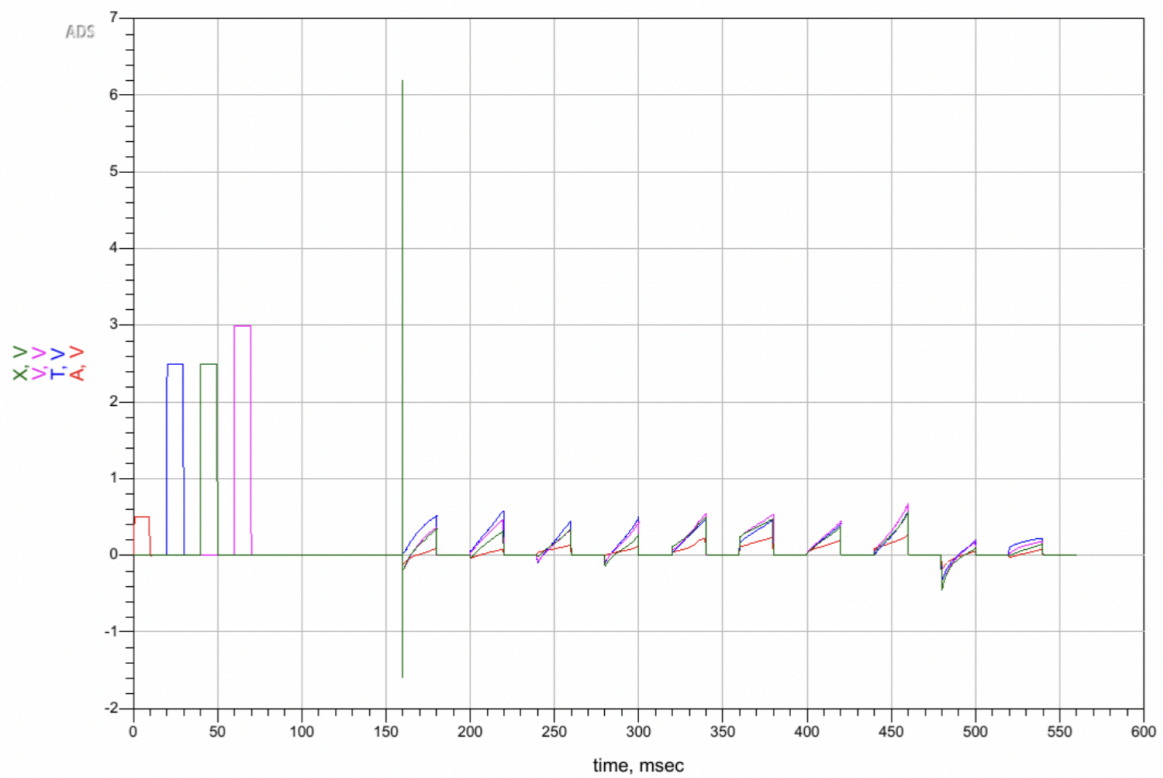


Fig. (TOP) T LETTER Training Data (10 images) as inputs to the 2-layer FFNN

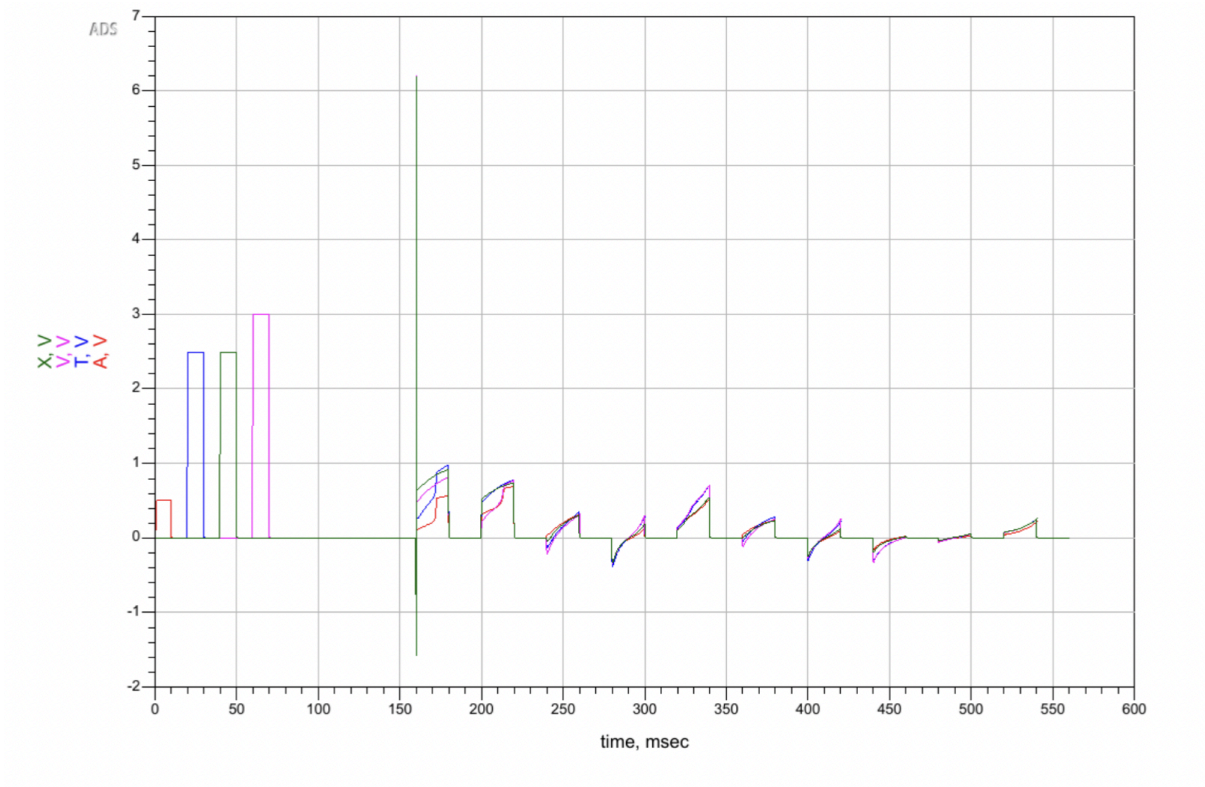


Fig. (TOP) V LETTER Training Data (10 images) as inputs to the 2-layer FFNN

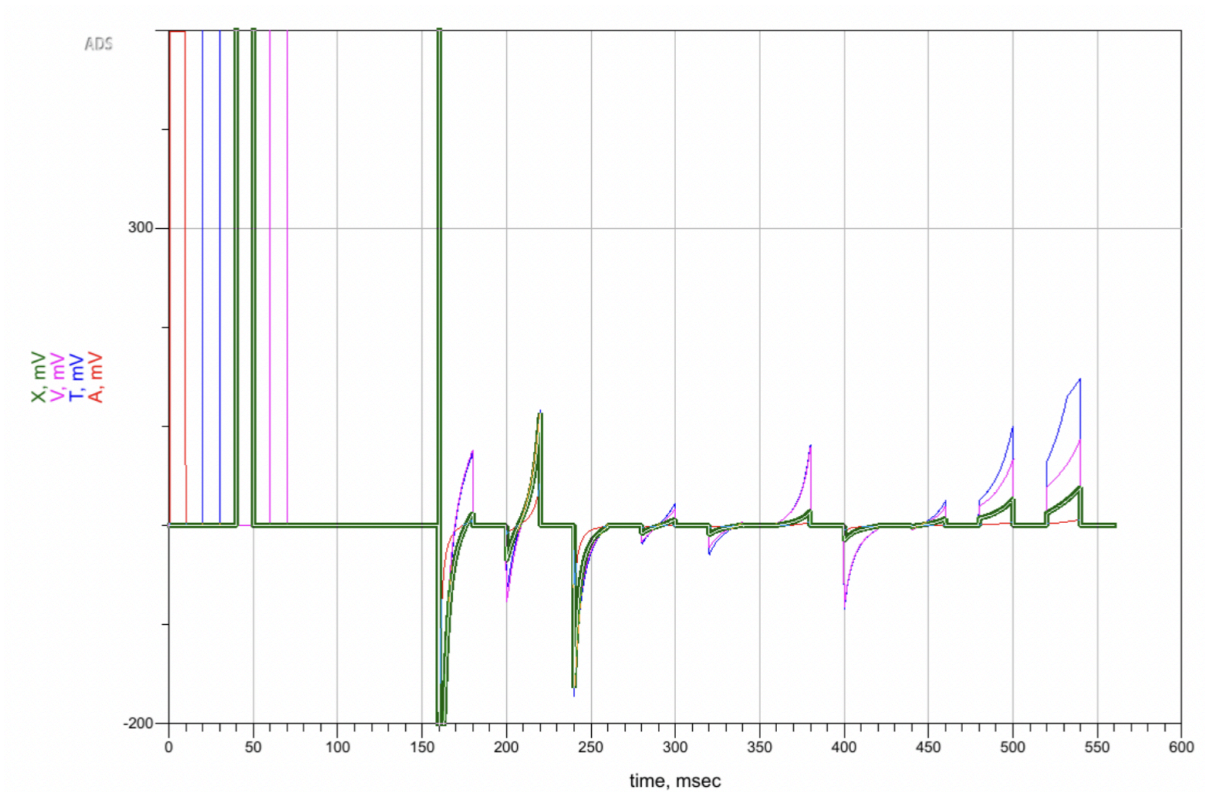


Fig. (TOP) X LETTER Training Data (10 images) as inputs to the 2-layer FFNN

References:

https://www.researchgate.net/publication/349211497_21_Multiplexer_Using_Different_Design_Styles_Comparative_Analysis/figures?lo=1

https://www.researchgate.net/publication/273067439_A_neuromorphic_hardware_framework_based_on_population_coding/figures?lo=1

https://www.researchgate.net/publication/352070805_Fusion_of_Fully_Integrated_Analog_Machine_Learning_Classifier_with_Electronic_Medical_Records_for_Real-Time_Prediction_of_Sepsis_Onset/figures?lo=1&utm_source=google&utm_medium=organic

https://www.researchgate.net/publication/327425781_Design_and_analysis_of_high-speed_8-bit_ALU_using_18_nm_FinFET_technology/figures?lo=1

<https://electronics.stackexchange.com/questions/657902/what-are-the-activation-functions-that-can-be-generated-using-op-amps-and-filter>

https://www.researchgate.net/figure/Mux-using-Transmission-Gates_fig3_303370654